

Ratio and Proportion

Ratio and Proportion. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner works with ratio as a multiplicative relationship — scaling recipes, building ratio tables, recognising equivalent ratios, and sharing quantities fairly. The through-line is that quantities in a ratio all change by the **same factor**, and that ratio is about multiplying, not adding.

Driving Question: *Can maths help us share, scale and sustain things fairly?*

The mathematics, in plain terms

A ratio compares quantities, e.g. 1 : 2. To scale, multiply every part by the same number (double 1 : 2 to 2 : 4 — an equivalent ratio). To share an amount in a ratio, add the parts to find the total number of equal shares, find the size of one share, then give each part its number of shares. The single biggest idea is **multiplicative** thinking: the parts scale by a common factor; they are not changed by adding a fixed amount.

Multiplying, not adding — the heart of ratio

The most common and important error in ratio is **additive** thinking: changing 1 : 2 to 2 : 3 by adding one to each part, instead of to 2 : 4 by doubling. Whenever a learner adjusts a ratio, it is worth asking “what did you multiply each part by?” — because a single common multiplier is what keeps a ratio (and a recipe) true. Ratio tables make this visible: the same multiplier moves you across every row.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	A ratio connects quantities: change one and the others must change with it by the same factor.	~15 min
Movement	Scaling recipes up and down — watching every ingredient move by the same multiplier.	~12 min
Reflection	Asking: did I multiply every part by the same number? Is this ratio in its simplest form?	~7 min
Creativity	Designing your own recipe or model and scaling it for a different number of people.	~4 min
Rest	The pause to predict amounts before revealing them.	~3 min

Nutrition	Contextual (Mode A): this lesson lives in food, recipes and fair sharing — the mathematics of feeding people and sharing resources sustainably.	~4 min
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Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Additive instead of multiplicative thinking

Changing $1 : 2$ to $2 : 3$ by adding one to each part. Repair: “What did you multiply each part by? A ratio keeps its taste only if every part scales by the same number.”

Losing the order of the ratio

Writing the parts in a different order for the answer. Repair: “Which quantity does each number stand for? Keep them in the same order throughout.”

Confusing part-to-part with part-to-whole

When sharing, using a part as if it were the whole. Repair: “How many equal shares are there altogether? Add the parts of the ratio first.”

Not simplifying, or not seeing equivalence

Repair: “Is there a number that divides both parts? Two ratios are equivalent if one scales to the other.”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: scaling by a multiplier, equivalent fractions, and sharing into parts. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the multiplier in view:

- “Going from this column to that one, what did you multiply by?”
- “Does that same multiplier act on every ingredient?”

- “To share fairly, how many equal parts are there in total?”
- “Is this ratio in its simplest form — can both parts be divided by the same number?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“They changed the ratio by adding — is that wrong?”

Usually, yes — ratio scales by multiplying. Ask what single number they multiplied each part by; if they added a fixed amount, the relationship (and the recipe) breaks.

“Do fractions and ratios overlap?”

Closely. Equivalent fractions and equivalent ratios are the same idea — the same relationship written with different numbers.

“Why all the recipes?”

Recipes make ratio real and show why fair, proportional scaling matters — the same maths shares food, budgets and resources sustainably.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.
Developing	Carries out the method with prompts; can explain part of the reasoning.
Secure	Works through independently and explains why each step is true.
Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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